

Muhammad Abbas

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EXPERIENCE

- Data Engineer I** | *Logiciel Services LLC.* — United States Full-time | May 2024 — Present
- Led the development of Lake & ELT pipelines (Cortex) and participated in the architecture meeting, convincing the team to use Spark instead of Nifi for ELT along with Kafka instead of Pulsar.
 - Developed Rust pipelines (Ferroflow) for EOD processing of FIX logs along with Nifi pipeline, speeding up the legacy .NET utility process by 10x, saving 400 minutes daily of AWS EC2 instance.
 - Developed a finetuned SQL-query chatbot for financial logs, involving collaborating with FinTech teams and Global Operation teams to ensure the product fits the use-case, smoothening the process for business teams by 40%.
- Junior Data Engineer** | *ImproData* — Victoria, Australia Part-time | Dec 2023 – March 2024
- Worked on Azure Databricks (PySpark) for risk management of cyber-security data, building bronze, silver and gold layers.
 - Worked on automated pipeline dev using GCP Cloud Functions to transfer compressed json bulk data from local to GCS & population to BigQuery
- DevOps Intern** | *Pullflow Inc.* — San Jose, California Full-time | July 2023 - Aug 2023
- Worked on AWS EKS (Kubernetes) and Docker. Was responsible for designing & implementing the infrastructure that was on AWS to on-premise server.
 - Worked on Budibase Dashboarding & hosting it on pullflow subdomain using traefik for loadbalancing.
- Data Engineering Intern** | *United Bank Limited* — Karachi, Pakistan Full-time | May 2023 – Jun 2023
- Built an Informatica pipeline for streaming UBL's data from Qatar & Bahrain into Hive Data Lake; participated in architecture analysis to ensure production AI pipelines on UBL Pak's data work for Qatar & Bahrain.
- Student Mentor** | *Habib University* — Karachi, Pakistan Part-time | Aug 2023 – May 2024
- *Teaching Assistant for Object Oriented Programming*: Took student labs and helped them with project & assignments.
 - *Git & Github Workshop Instructor*: Conducted two workshops on Git and GitHub for beginner and intermediate users.
 - *Teaching Assistant for Data Structures & Algorithms*: Assisted students during labs and with assignments DSA.

EDUCATION

Habib University **Bachelors' in Computer Science** | GPA : 3.31 August 2020 – June 2024

PROJECTS

- Cortex: Data Lake & ELT Data Pipelines (Logiciel)** | *Spark, Airflow, Kafka, Trino, AWS Cloud, PostgreSQL* Jan 2025 - March 2025
- Develop Spark pipelines using Scala for data ingestion and transformation of 375 Fix sessions daily spanning multiple products and environments, Orchestrate the workflow using Airflow dynamic dags.
 - Setup Datalake and Mesh architecture S3 along with Trino for federated query. Utilize AWS Glue for cataloging
- Yohsin: High-Performance Object Serialization Tool (Hackathon)** | *Rust, tokio* 09 Jan 2025
- A high-performance, multi-threaded yet fault-tolerant, zero-copy serializer/deserializer handling serialization of 1M records (2.25GB) per second, supporting range deserialization, random access and checkpointing for stateful recovery, utilizing memory-mapped files and async I/O for quicker operations with no memory overhead.
- Astroflow: Bayesian Flow Networks Based Model (FYP)** | *PyTorch, Django, Postgres, SQLite, Javascript* Aug 2023 - May 2024
- A Deep Learning framework to estimate population parameters of Binary Black holes under the leadership of RIT instructor, the framework was based on Normalizing flows for inference and Neural Nets for summarization and Feature extraction from unseen data.
- Ferroflow: ETL framework (Logiciel)** | *Rust* June 2024 - July 2024
- A Rust framework inspired by Nifi for ETL, modularized with processors and pipelines to facilitate re-usability.
 - Improved the Large-File-Sorting, chunking and processing by 10x using Ferroflow and Nifi, sorting a massive FIX log file in 50 mins that took 500 minutes with .NET utility daily on Cloud.
- Metosis: Fix Logs Chatbot (Logiciel)** | *Python, langChain, langGraph, SQL, Trino* Sept 2024 - Jan 2025
- A multi-agentic chatbot that is fine-tuned on Fix logs and FinTech concepts, queries on our data-lake using a Trino's SQL-like interface in real-time. Enabled business teams to obtain insights directly, making the process 40% faster.

CERTIFICATIONS AND ACHIEVEMENTS

- IELTS-Academic** : 8 out of 9 **GRE Math** : 165 out of 170
- HU-TOPS Scholarship**: 100% Merit Scholarship for 4-year Bachelor's degree in Computer Science
- ICPC Regionals**: Secured 19th position in ICPC Regionals Asia Topi 2022
- Game of Codes 5.0** Winner of Competitive Programming Competition Game of Codes 5.0
- Programming Certifications**:
- Building Data Lakes on AWS - [AWS - Coursera]
 - Apache Spark 3 with Scala Hands-on with Big Data - [SkillShare]
 - ETL and Data Pipelines with Shell, Airflow and Kafka - [IBM - Coursera]

TECHNICAL SKILLS

- Languages** : Python, Rust, Scala, C/C++, Golang, SQL, NoSQL, HTML, CSS, JS (basic)
- Data Eng. Tools** : Spark, Airflow, Kafka, Trino/Presto/Athena, AWS Glue, Iceberg, Hive, Hadoop, Nifi, Databricks, Informatica
- AI Frameworks** : Tensorflow, PyTorch, Langchain, LangGraph, Pydantic-AI, JAX, NumPyro
- Software Frameworks & Databases** : Django, Flutter, REST, Qdrant, Milvus, Neo4j, PostgreSQL, Amazon RDS, MongoDB
- Cloud, DevOps and API Tools** : Linux (Longtime Desktop User), AWS, Azure Cloud, GCP, Git, Docker, Kubernetes, Postman,
- Others** : Polars, Pandas, Design Patterns, Debugging, Root Cause Analysis, Architecture Planning, Working with JSON & Parquet